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Applications of Differentiation

Stationary means the first derivative is zero. Its sign says up or down; the second derivative says maximum or minimum.

THE TOPIC ON ONE MAP

Applications of Differentiation

01 – Increasing, decreasing

$f'(x) > 0$ $f'(x) < 0$ for all x

02 – Stationary points

$f'(x) = 0$ f'' : max or min show that $k = \dots$

03 – Optimisation

one variable only max / min value justify with f''

04 – With areas

stationary point first then integrate

THE TWO DERIVATIVES

$$\frac{dy}{dx} = 0 \text{ stationary}$$

$$\frac{d^2y}{dx^2} < 0 \text{ maximum}$$

$$\frac{d^2y}{dx^2} > 0 \text{ minimum}$$

$$\frac{d^2y}{dx^2} > 0$$

rewrite first: $\sqrt{x} = x^{\frac{1}{2}}$, $\frac{1}{x^2} = x^{-2}$ · increasing: $\frac{dy}{dx} > 0$ decreasing: $\frac{dy}{dx} < 0$

01 Increasing and Decreasing Functions

Differentiate, then solve the inequality $f'(x) > 0$ (increasing) or $f'(x) < 0$ (decreasing). For a quadratic $f'(x)$, find its roots and sketch it. “Increasing for all x ”: show $f'(x)$ is always positive — complete the square.

WHEN YOU SEE → YOU DO

“find the values of x for which y is increasing”

→ $f'(x) > 0$: roots, sketch, read

“state the range of values of x for which y is decreasing”

→ $f'(x) < 0$

“show that f is increasing for all x ”

→ $f'(x) = a(x + p)^2 + q$ with $a, q > 0$

“prove that C has no turning points”

→ $f'(x) = 0$ has no real roots: $b^2 - 4ac < 0$

WORKED MODEL · OCTOBER 2019

$y = 2x^2(x - 5)$. Stationary points and where y is increasing.

1 $y = 2x^3 - 10x^2$, $\frac{dy}{dx} = 6x^2 - 20x = 2x(3x - 10)$

expand, differentiate

2 $\frac{dy}{dx} = 0$: $x = 0$, $\frac{10}{3}$

stationary

ANSWER increasing for $x < 0$ or $x > \frac{10}{3}$

IDEA 01

Practise: Increasing and Decreasing Functions

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Where is it increasing?

P2 · TOPIC 11 3 marks

Find the values of x for which $f(x) = x^3 - 6x^2 + 5$ is increasing.

Solve $f'(x) > 0$: sketch the quadratic.

Q2 **DR ESLAM** Increasing for all x

P2 · TOPIC 11 3 marks

Show that $f(x) = x^3 + 3x^2 + 6x$ is increasing for all x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 11 6 marks

1 [i] Find an expression for $f'(x)$ when $f(x) = x^3 + x^2 - 5x$.

[ii] Solve the equation $3x^2 + 2x - 5 = 0$

[iii] Hence, or otherwise, find the values of x for which $f(x)$ is a decreasing function.

(6 marks)**Q4****PRACTICE**

Practice · Hard

P2 · TOPIC 11 5 marks

1 Find the values of x for which $f(x) = x^3 - 5x^2 + 3x - 2$ is a decreasing function.

(5 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11**4 marks**

- 2 Show that the function $f(x) = \sqrt{x} - \frac{7}{\sqrt{x}}$, $x > 0$, is increasing for all x in its domain.

(4 marks)**PAST PAPERS****Q6****PAST PAPER**

Pearson IAL WMA12 · October 2019 · Q1

P2 · TOPIC 11**6 marks**

1. A curve C has equation $y = 2x^2(x - 5)$
- (a) Find, using calculus, the x coordinates of the stationary points of C .
- (b) Hence find the values of x for which y is increasing.

(4)**(2)****(Total 6 marks)**

Q7 **PAST PAPER** Pearson IAL WMA12 · October 2019 · Q10**P2 · TOPIC 11** **7 marks**

10. The curve C has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where a and b are constants.

Given that

- the point $(2, 5)$ lies on C
- the gradient of the curve at $(2, 5)$ is 7

(a) find the value of a and the value of b .

(4)

(b) Prove that C has no turning points.

(3)

Q8**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q6

P2 · TOPIC 11 9 marks

6.

$$f(x) = 4x^3 + px^2 + 8x + q$$

where p and q are constants.

Given that

- $(2x + 3)$ is a factor of $f(x)$
- $f(x)$ has a remainder of -5 when divided by $(x + 2)$

(a) (i) show that $p = 10$

(ii) find the value of q .

(5)

(b) Hence find the range of values of x for which $f(x)$ is decreasing.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

02 Stationary Points and Their Nature: $\frac{d^2y}{dx^2}$

Set $\frac{dy}{dx} = 0$ and solve for x ; substitute into y for the coordinates. Then put each x into $\frac{d^2y}{dx^2}$: negative is a maximum, positive is a minimum.

WHEN YOU SEE → YOU DO

“find the coordinates of the stationary point(s)”

→ $y' = 0$, then y

“determine the nature”, “justify”

→ sign of y'' at that x

“ P with $x = \frac{1}{2}$ is a stationary point: show $k = \dots$ ”

→ $y'(\frac{1}{2}) = 0$

“show that the x coordinates satisfy $2x^3 - 10x^2 + \dots = 0$ ”

→ $y' = 0$, clear the powers

“one turning point is at $x = 1$: find the others”

→ factor $(x - 1)$ out of the cubic

Rewrite

Differentiate

Set = 0 and solve

Second derivative

WORKED MODEL • JANUARY 2022

$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20$. Stationary point and its nature.

1 $\frac{dy}{dx} = \frac{27}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 27 = 3x$

multiply by $2x^{\frac{1}{2}}$

2 $x = 9: y = 81 - 27 - 20 = 34$

coordinates

ANSWER $(9, 34), \frac{d^2y}{dx^2} < 0$: maximum

IDEA 02

Practise: Stationary Points and Their Nature:

$$\frac{d^2y}{dx^2}$$

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Find and classify

P2 · TOPIC 11**5 marks**

Find the stationary points of $y = 2x^3 - 3x^2 - 12x + 1$ and their nature.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 11 7 marks

2 (a) The curve C has equation $y = 3x^3 + 6x^2 - 5x + 1$.

Find expressions for $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(3 marks)

(b) [i] Evaluate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = \frac{1}{3}$.

[ii] What does your answer to part [b] tell you about curve C at the point where $x = \frac{1}{3}$?

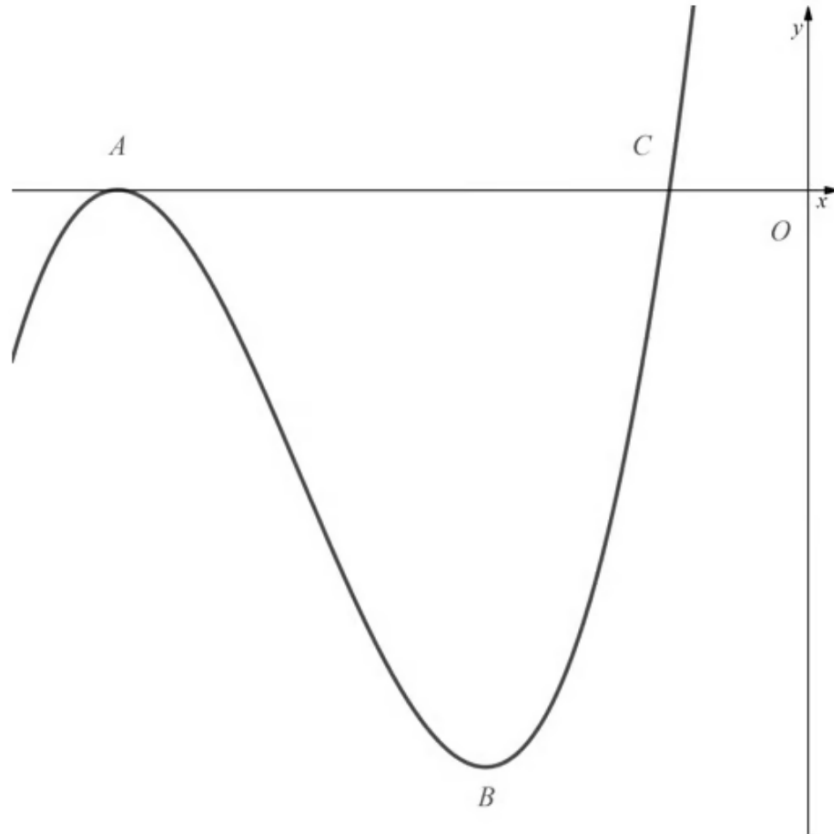
(4 marks)

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 11 7 marks

- 4 (a)** The diagram below shows part of the curve with equation $y = x^3 + 11x^2 + 35x + 25$. The curve touches the x -axis at A and cuts the x -axis at C . The points A and B are stationary points on the curve.



Using calculus, and showing all your working, find the coordinates of A and B .

(5 marks)

- (b)** Show that $(-1, 0)$ is a point on the curve and explain why those must be the coordinates of point C .

(2 marks)

Q12**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11 7 marks

3 (a) A curve is described by the equation $y = f(x)$, where $f(x) = 7 - 2x^2 + \sqrt{x}$, $x \geq 0$.

Find $f'(x)$ and $f''(x)$.

(3 marks)

(b) P is the stationary point on the curve.

Find the coordinates of P and determine its nature.

(4 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2020 · Q10

P2 · TOPIC 11 10 marks

10. A curve C has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where k is a constant.

The point P with x coordinate $\frac{1}{2}$ lies on C .

Given that P is a stationary point of C ,

(a) show that $k = -\frac{3}{2}$ (4)

(b) Determine the nature of the stationary point at P , justifying your answer. (2)

The curve C has a second stationary point.

(c) Using algebra, find the x coordinate of this second stationary point. (4)

Q14 **PAST PAPER** Pearson IAL WMA12 · January 2021 · Q2**P2 · TOPIC 11** **7 marks**

2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

- (a) Using calculus, find the x coordinates of the stationary points of the curve. (4)
- (b) Justify, by further calculus, the nature of all of the stationary points of the curve. (3)

(Total 7 marks)

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q2

P2 · TOPIC 11**8 marks**

2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving each term in simplest form. (2)
- (b) Hence find the coordinates of the stationary point of C . (4)
- (c) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of the stationary point of C . (2)

(Total 8 marks)

Q16**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q8

P2 · TOPIC 11**8 marks**

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A curve has equation

$$y = 256x^4 - 304x - 35 + \frac{27}{x^2} \quad x \neq 0$$

(a) Find $\frac{dy}{dx}$

(3)

(b) Hence find the coordinates of the stationary points of the curve.

(5)

(Total 8 marks)

Q17 PAST PAPER Pearson IAL WMA12 · October 2022 · Q7

P2 · TOPIC 11 9 marks

7. The curve C has equation

$$y = \frac{12x^3(x - 7) + 14x(13x - 15)}{21\sqrt{x}} \quad x > 0$$

(a) Write the equation of C in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where a , b , c and d are fully simplified constants.

(3)

The curve C has three turning points.

Using calculus,

(b) show that the x coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Given that the x coordinate of one of the turning points is 1(c) find, using algebra, the exact x coordinates of the other two turning points.*(Solutions based entirely on calculator technology are not acceptable.)*

(3)

(Total for Question 7 is 9 marks)

Q18 PAST PAPER Pearson IAL WMA12 · June 2023 · Q7

P2 · TOPIC 11 8 marks

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river, H metres, t days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \quad 0 \leq t \leq 7$$

Given that H has a stationary value at $t = \alpha$

- (a) use calculus to show that α satisfies the equation

$$5\alpha^2 - 18\alpha - 20 = 0 \quad (5)$$

- (b) Hence find the value of α , giving your answer to 3 decimal places. (1)

- (c) Use further calculus to prove that H is a maximum at this value of α . (2)

(Total for Question 7 is 8 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q4

P2 · TOPIC 11 7 marks

4. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ giving each term in simplest form. (2)
- (b) Hence find the x coordinate of the stationary point of C . (2)
- (c) (i) Find $\frac{d^2y}{dx^2}$ giving each term in simplest form.
- (ii) Hence determine the nature of the stationary point of C , giving a reason for your answer. (2)
- (d) State the range of values of x for which y is decreasing. (1)

(Total for Question 4 is 7 marks)

03 Optimisation: Build the Formula, Then Differentiate

A real object with a fixed volume (or area, or perimeter) and two unknown lengths. Use the fixed fact to write one length in terms of the other, substitute, and get a formula in one variable. Then $\frac{d}{dx} = 0$, and justify the minimum with the second derivative.

WHEN YOU SEE → YOU DO

“the volume is 972 cm^3 ”

→ height in terms of x

“show that $S = 6x^2 + \frac{2592}{x}$ ”

→ surface area, substitute the height

“open-topped”

→ leave out the top face

“use calculus to find the minimum value”

→ $S' = 0$, then S at that x

“justify that this is a minimum”

→ $S'' > 0$ at that x

WORKED MODEL · JANUARY 2023

$S = 6x^2 + \frac{2592}{x}$. Find the minimum surface area.

1 $S' = 12x - \frac{2592}{x^2} = 0 \Rightarrow x^3 = 216$

stationary

2 $x = 6; S'' = 12 + \frac{5184}{x^3} > 0$

minimum

ANSWER $S = 216 + 432 = 648 \text{ cm}^2$

× **TRAP** Differentiating before the formula has only one variable. Eliminate first.

IDEA 03

Practise: Optimisation: Build the Formula, Then Differentiate

My questions first, then selected practice, then real past papers.

Q20 DR ESLAM A box

P2 · TOPIC 11 5 marks

An open box has a square base of side x cm and volume 500 cm^3 . Show $A = x^2 + \frac{2000}{x}$ and find the minimum area.

Height from the volume, then the area in x only.

SELECTED PRACTICE

Q21**PRACTICE**

Practice · Medium

P2 · TOPIC 11 6 marks

- 5 (a)** A company manufactures food tins in the shape of cylinders which must have a constant volume of $150\pi \text{ cm}^3$. To lessen material costs the company would like to minimise the surface area of the tins.

By first expressing the height h of the tin in terms of its radius r , show that the surface area of the cylinder is given by $S = 2\pi r^2 + \frac{300\pi}{r}$.

(2 marks)

- (b)** Use calculus to find the minimum value for the surface area of the tins. Give your answer correct to 2 decimal places.

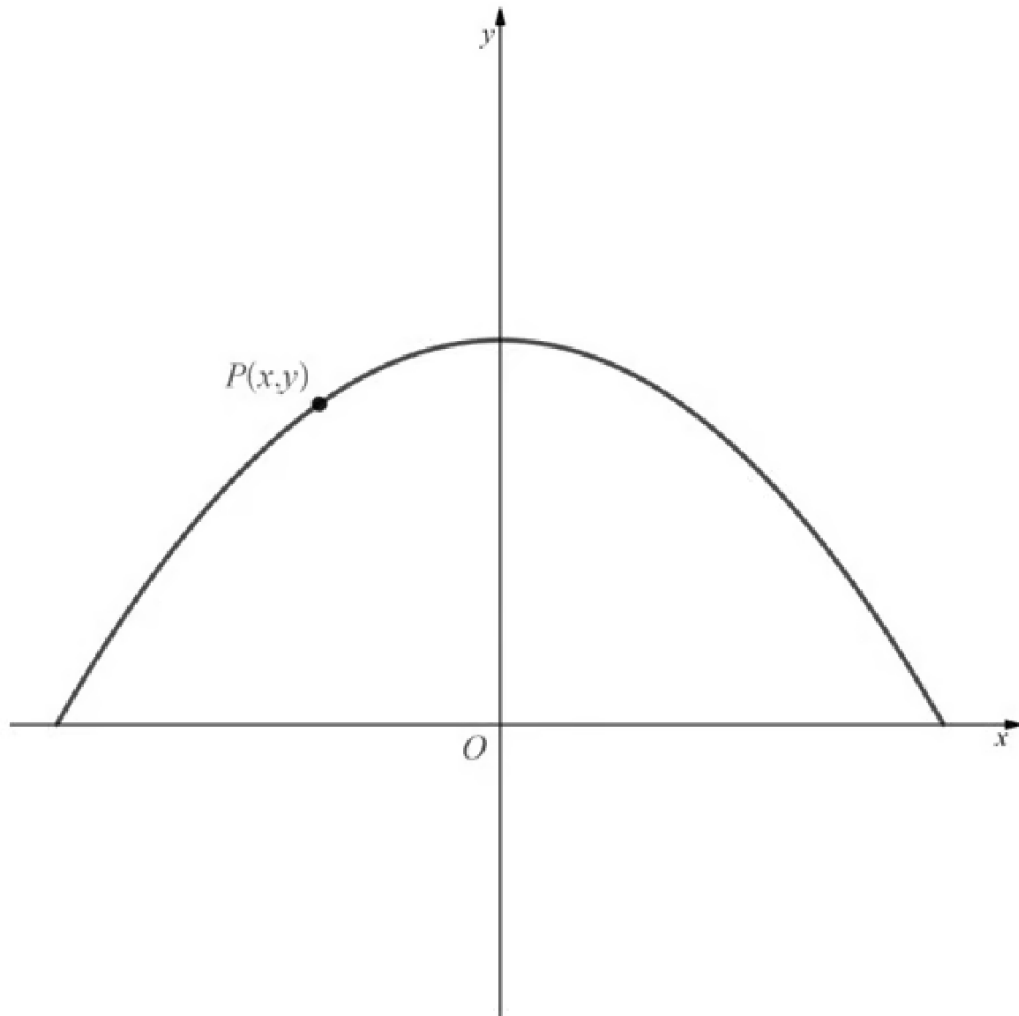
(4 marks)

Q22**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11 11 marks

- 4 (a)** The diagram below shows the part of the curve with equation $y = 3 - \frac{1}{4}x^2$ for which $y > 0$. The marked point $P(x, y)$ lies on the curve. O is the origin.



Show that $OP^2 = 9 - \frac{1}{2}x^2 + \frac{1}{16}x^4$.

(3 marks)

- (b)** Find the minimum distance from O to the curve, using calculus to prove that your answer is indeed a minimum.

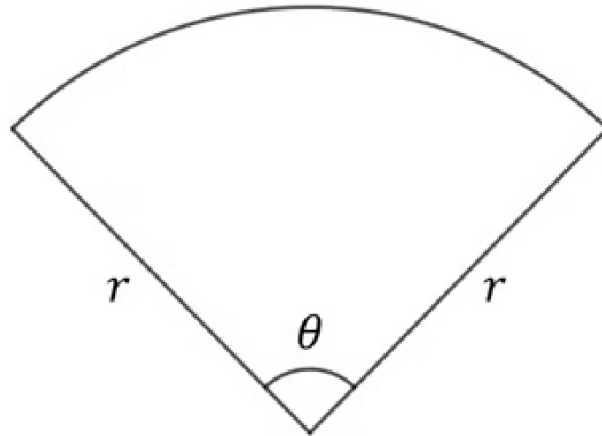
(8 marks)

Q23**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11 9 marks

- 5 (a) The top of a patio table is to be made in the shape of a sector of a circle with radius r and central angle θ , where $0^\circ < \theta < 360^\circ$.



Although r and θ may be varied, it is necessary that the table have a fixed area of $A \text{ m}^2$.

Explain why $r > \sqrt{\frac{A}{\pi}}$.

(2 marks)

- (b) Show that the perimeter, P , of the table top is given by the formula

$$P = 2r + \frac{2A}{r}$$

(2 marks)

- (c) Show that the minimum possible value for P is equal to the perimeter of a square with area A . Be sure to prove that your value is a minimum.

(5 marks)

PAST PAPERS

Q24 PAST PAPER Pearson IAL WMA12 · June 2021 · Q9

P2 · TOPIC 11 10 marks

9.

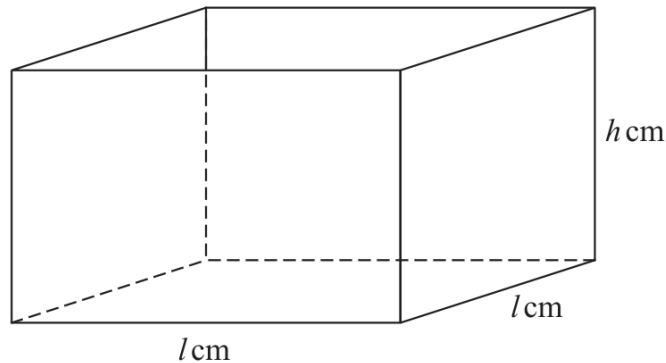


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is h cm, and the base edges each have length l cm.

Given that the volume of the box is $250\,000\text{ cm}^3$

(a) show that the external surface area, $S\text{ cm}^2$, of the box is given by

$$S = \frac{250\,000}{h} + 2000\sqrt{h} \quad (3)$$

(b) Use algebraic differentiation to show that S has a stationary point when $h = 250^k$ where k is a rational constant to be found.

(5)

(c) Justify by further differentiation that this value of h gives the minimum external surface area of the box.

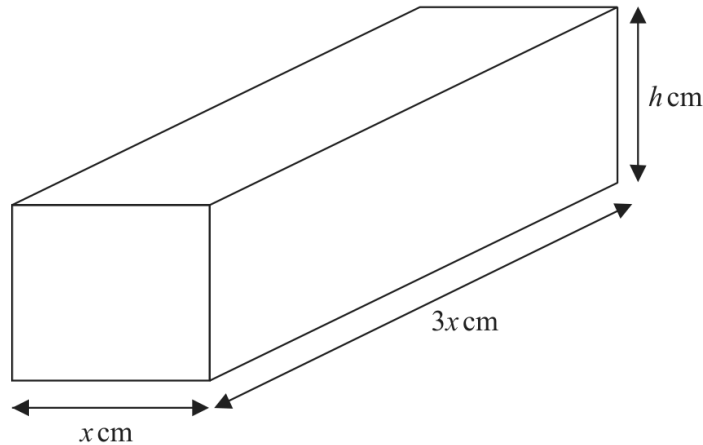
(2)

Q25**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q2

P2 · TOPIC 11 9 marks

2.

In this question you must show all stages of your working.**Solutions based entirely on calculator technology are not acceptable.****Figure 2**

A brick is in the shape of a cuboid with width $x \text{ cm}$, length $3x \text{ cm}$ and height $h \text{ cm}$, as shown in Figure 2.

The volume of the brick is 972 cm^3

(a) Show that the surface area of the brick, $S \text{ cm}^2$, is given by

$$S = 6x^2 + \frac{2592}{x} \quad (3)$$

(b) Find $\frac{dS}{dx}$ (1)

(c) Hence find the value of x for which S is stationary. (2)

(d) Find $\frac{d^2S}{dx^2}$ and hence show that the value of x found in part (c) gives the minimum value of S . (2)

(e) Hence find the minimum surface area of the brick. (1)

(Total for Question 2 is 9 marks)

Q26 PAST PAPER Pearson IAL WMA12 · January 2025 · Q3

P2 · TOPIC 11 10 marks

3.

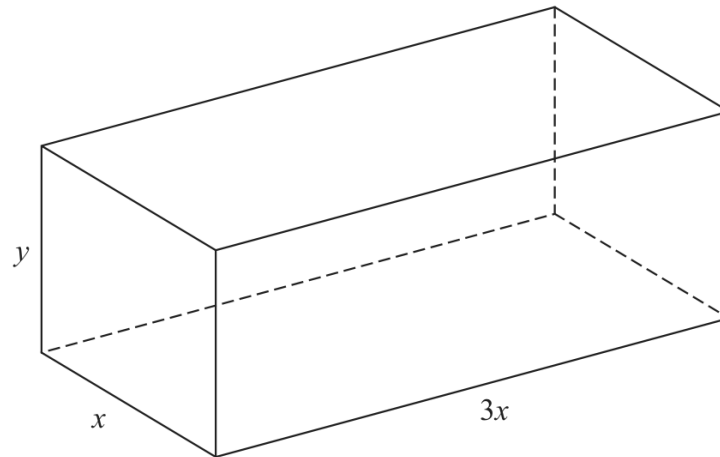
**Figure 1**

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle $3x$ metres by x metres.

The height of the container is y metres as shown in Figure 1.

Given that the capacity of the container is 120m^3

(a) show that the area $A\text{m}^2$ of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where P and Q are positive constants to be found.

(4)

(b) Use calculus to find the value of x for which A has a stationary value, giving your answer to 3 significant figures.

(4)

(c) Find $\frac{d^2A}{dx^2}$ and hence show that the value of x found in part (b) gives the minimum value of A .

(2)

(Total for Question 3 is 10 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q9

P2 · TOPIC 11 11 marks

9:

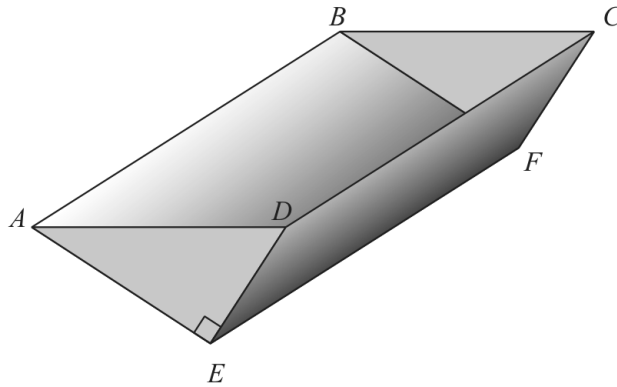
**Figure 2**

Figure 2 shows a sketch of an open container.

The sides $CDEF$ and $ABFE$ are rectangles.

The ends ADE and BCF are congruent (identical) right-angled triangles.

The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$ metres
- $DE = CF = 2x$ metres
- $AB = DC = EF = L$ metres

and the capacity of the container is 12 m^3

(a) show that the area of metal used to make the container, $S \text{ m}^2$, is given by

$$S = Px^2 + \frac{Q}{x}$$

where P and Q are positive integers to be found.

(4)

Given that x can vary,

(b) use algebraic calculus to find the minimum value of S , giving your answer to one decimal place.

(5)

(c) Justify that the value of S found in part (b) is a minimum.

(2)

(Total for Question 9 is 11 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q9

P2 · TOPIC 11 11 marks

9.

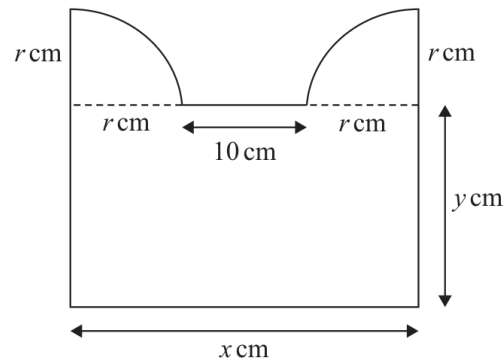
**Figure 4**

Figure 4 shows a sketch of a design for a logo.

The design consists of a rectangle of base x cm and height y cm with two quarter circles of equal radius r cm joined to the top edge with a gap of 10 cm between the nearest point of each quarter circle, as shown in Figure 4.

(a) Show that the area, A cm², of the design is given by

$$A = xy + \frac{\pi}{8}(x - 10)^2 \quad (2)$$

Given that

- the total area of the design is one square metre
- the perimeter, P cm, of the design is given by $P = 2y + \left(2 + \frac{\pi}{2}\right)x - 5\pi$

(b) show that

$$P = \frac{20000 - 25\pi}{x} + \left(2 + \frac{\pi}{4}\right)x \quad (4)$$

(c) Use algebraic differentiation to find the value of x that gives the minimum value for P , giving the answer to 3 significant figures.

(3)

By considering $\frac{d^2P}{dx^2}$

(d) justify that the value for x found in part (c) gives the minimum value for P .

(2)

(Total for Question 9 is 11 marks)

04 Stationary Points with Areas

These questions join this topic to Integration: find the stationary point first (it gives a limit or the height of a line), then integrate between the right limits. Areas below the x -axis come out negative.

WHEN YOU SEE → YOU DO

“the line l passes through M and is parallel to the x -axis”

→ y = the y of the stationary point

“the region bounded by C , l and the y -axis”

→ line minus curve, or curve minus line

“area of R_1 equals area of R_2 ”

→ $\int_0^k y \, dx = 0$

“use algebraic integration”

→ show the integral and the substitution

WORKED MODEL · JUNE 2024

$y = 2x^{\frac{3}{2}}(4 - x)$. R_1 above the axis and R_2 below it, with equal areas. Find k .

$$1 \quad \int_0^k (8x^{\frac{3}{2}} - 2x^{\frac{5}{2}}) \, dx = \frac{16}{5}k^{\frac{5}{2}} - \frac{4}{7}k^{\frac{7}{2}} = 0$$

equal areas: total 0

ANSWER $k = \frac{28}{5}$

IDEA 04

Practise: Stationary Points with Areas

My questions first, then real past papers.

Q29**DR ESLAM**

Then the area

P2 · TOPIC 11**4 marks**

$y = x(4 - x)$. Find the stationary point and the area between the curve and the x -axis.

PAST PAPERS

Q30 PAST PAPER Pearson IAL WMA12 · October 2021 · Q8

P2 · TOPIC 11 10 marks

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

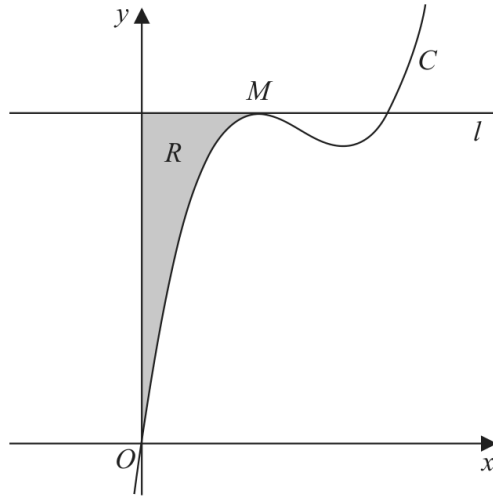


Figure 2

Figure 2 shows a sketch of part of the curve C with equation

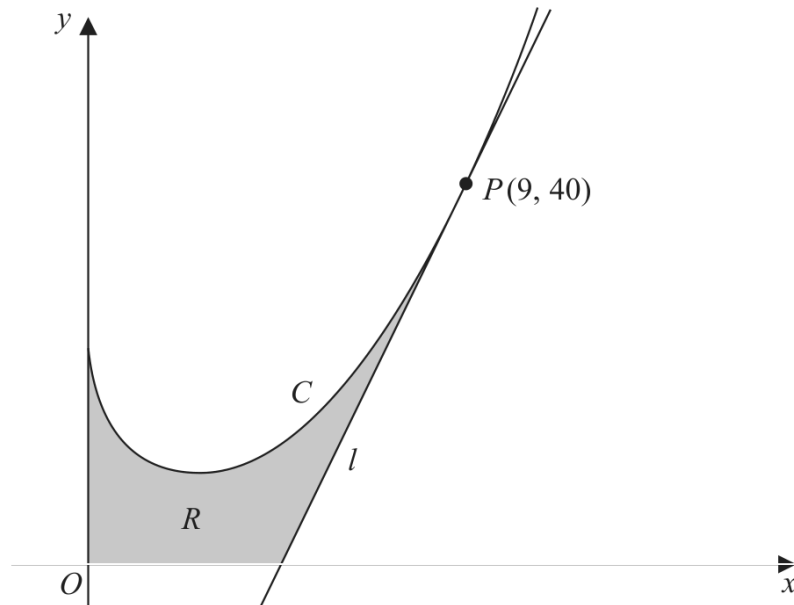
$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point M is the maximum turning point of C and is shown in Figure 2.Given that the x coordinate of M is 2(a) show that $k = 28$ (3)(b) Determine the range of values of x for which y is increasing. (2)The line l passes through M and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve C , the line l and the y -axis.(c) Find, by algebraic integration, the exact area of R . (5)**(Total 10 marks)**

Q31 PAST PAPER Pearson IAL WMA12 · October 2023 · Q9

P2 · TOPIC 11 12 marks

9.

**Figure 3**

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = \frac{2}{3}x^2 - 9\sqrt{x} + 13 \quad x \geq 0$$

(a) Find, using calculus, the range of values of x for which y is increasing.

(4)

The point P lies on C and has coordinates $(9, 40)$.

The line l is the tangent to C at the point P .

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the line l , the x -axis and the y -axis.

(b) Find, using calculus, the exact area of R .

(8)

(Total for Question 9 is 12 marks)

Q32**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q10

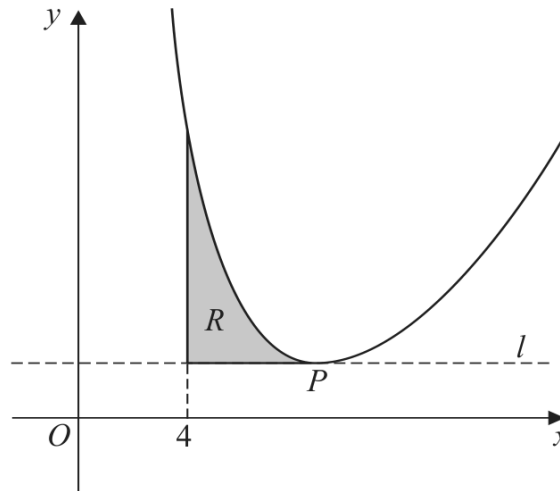
P2 · TOPIC 11 9 marks**10.****In this question you must show detailed reasoning.****Solutions relying entirely on calculator technology are not acceptable.****Figure 2**

Figure 2 shows a sketch of the curve with equation

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

The point P is the only stationary point on the curve.(a) Use calculus to show that the x coordinate of P is 9**(4)**The line l passes through the point P and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve, the line l and the line with equation $x = 4$ (b) Use algebraic integration to find the exact area of R .**(5)****(Total for Question 10 is 9 marks)**

Q33**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q9

P2 · TOPIC 11 8 marks

9.

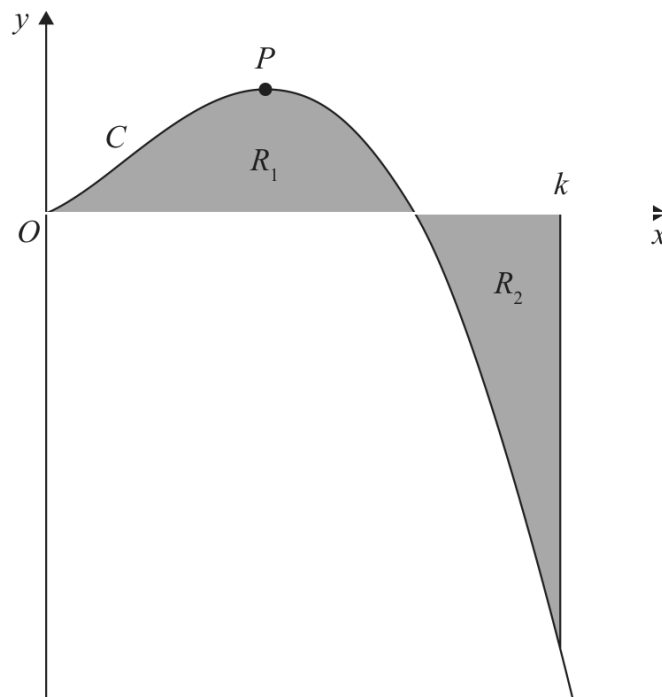
**Figure 1**

Figure 1 is a sketch of the curve C with equation

$$y = 2x^{\frac{3}{2}}(4 - x) \quad x \geq 0$$

The point P is the stationary point of C .

(a) Find, using calculus, the x coordinate of P .

(4)

The region R_1 , shown shaded in Figure 1, is bounded by C and the x -axis.

The region R_2 , also shown shaded in Figure 1, is bounded by C , the x -axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) find, using calculus, the exact value of k .

(4)

(Total for Question 9 is 8 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June (R paper) 2024 · Q10

P2 · TOPIC 11 9 marks

10.

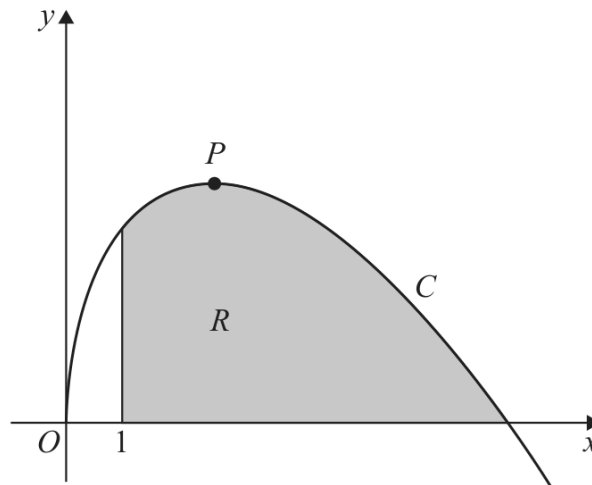


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{9x - x^2}{2\sqrt{x}} \quad x > 0$$

The point P is the stationary point on C .

(a) Find, using calculus, the x coordinate of P .

(4)

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 1$

(b) Using calculus, calculate the exact area of R .

(5)

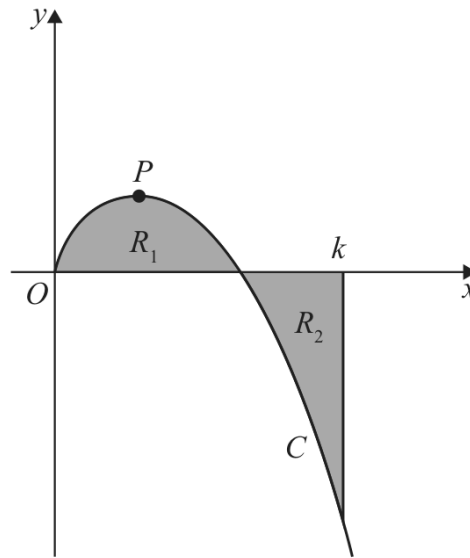
(Total for Question 10 is 9 marks)

Q35**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q10

P2 · TOPIC 11 8 marks

10.

**Figure 2**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 2 shows a sketch of the curve C with equation

$$y = \frac{\sqrt{x}(100 - x^2)}{40} \quad x \geq 0$$

The point P is a stationary point on C .

(a) Use algebraic differentiation to find the exact x coordinate of P .

(4)

The region R_1 , shown shaded in Figure 2, is bounded by C and the x -axis.

The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) use algebraic integration to find the exact value of k .

(4)

(Total for Question 10 is 8 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------|---|
| 01 | Increasing | $f' > 0$; decreasing $f' < 0$; “for all x ”: complete the square. |
| 02 | Stationary | $f' = 0$; $f'' < 0$ max, $f'' > 0$ min. |
| 03 | Optimisation | One variable first; $f' = 0$; justify with f'' . |
| 04 | With areas | Stationary point first, then integrate. |

BOTTOM LINE Rewrite $\rightarrow$ differentiate $\rightarrow = 0 \rightarrow$ second derivative decides max or min.

Answers

Applications of Differentiation · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · INCREASING AND DECREASING FUNCTIONS

- Q1** Dr Eslam · Where is it increasing?
 $f'(x) = 3x(x - 4) > 0: x < 0 \text{ or } x > 4$
- Q2** Dr Eslam · Increasing for all x
 $f'(x) = 3(x + 1)^2 + 3 > 0$
- Q3** Practice · Easy
 (i) $3x^2 + 2x - 5$ (ii) $x = -\frac{5}{3}, 1$ (iii) $-\frac{5}{3} < x < 1$
- Q4** Practice · Hard
 $f'(x) = 3x^2 - 10x + 3 < 0: \frac{1}{3} < x < 3$
- Q5** Practice · Very Hard
 $f'(x) = \frac{1}{2}x^{-\frac{1}{2}} + \frac{7}{2}x^{-\frac{3}{2}} = \frac{x+7}{2x^{3/2}} > 0 \text{ for } x > 0$
- Q6** Pearson IAL WMA12 · October 2019 · Q1
 (a) $x = 0, \frac{10}{3}$. (b) $x < 0 \text{ or } x > \frac{10}{3}$
- Q7** Pearson IAL WMA12 · October 2019 · Q10
 (a) $a = \frac{4}{3}, b = \frac{1}{3}$. (b) Proven: C has no turning points
- Q8** Pearson IAL WMA12 · June (R paper) 2024 · Q6
 (a)(i) $p = 10$. (a)(ii) $q = 3$. (b) $-1 < x < -\frac{2}{3}$

IDEA 02 · STATIONARY POINTS AND THEIR NATURE: $\frac{d^2y}{dx^2}$

- Q9** Dr Eslam · Find and classify
 $x = -1: (-1, 8) \max (y'' = -18); x = 2: (2, -19) \min (y'' = 18)$
- Q10** Practice · Easy
 (a) $9x^2 + 12x - 5, 18x + 12$ (b)(i) 0 and 18 (ii) a stationary point, a minimum
- Q11** Practice · Medium
 (a) $A(-5, 0)$ maximum, $B\left(-\frac{7}{3}, -\frac{256}{27}\right)$ minimum
 (b) $y(-1) = 0$; the curve crosses the axis only once more
- Q12** Practice · Very Hard
 (a) $f'(x) = -4x + \frac{1}{2}x^{-\frac{1}{2}}, f''(x) = -4 - \frac{1}{4}x^{-\frac{3}{2}}$ (b) $P\left(\frac{1}{4}, \frac{59}{8}\right)$, maximum ($f'' < 0$)
- Q13** Pearson IAL WMA12 · January 2020 · Q10
 (a) $k = -\frac{3}{2}$. (b) local maximum. (c) $x = \frac{1}{\sqrt{2}}$
- Q14** Pearson IAL WMA12 · January 2021 · Q2
 (a) $x = -2, \frac{8}{3}$. (b) local maximum at $x = -2$ and local minimum at $x = \frac{8}{3}$
- Q15** Pearson IAL WMA12 · January 2022 · Q2
 (a) $\frac{dy}{dx} = \frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$. (b) (9, 34). (c) local maximum
- Q16** Pearson IAL WMA12 · June 2022 · Q8
 (a) $\frac{dy}{dx} = 1024x^3 - 304 - \frac{54}{x^3}$. (b) $\left(-\frac{1}{2}, 241\right), \left(\frac{3}{4}, -134\right)$
- Q17** Pearson IAL WMA12 · October 2022 · Q7
 (a) $y = \frac{4}{7}x^{7/2} - 4x^{5/2} + \frac{26}{3}x^{3/2} - 10x^{1/2}$. (b) Proven. (c) $x = 2 - \frac{\sqrt{6}}{2}$ and $x = 2 + \frac{\sqrt{6}}{2}$
- Q18** Pearson IAL WMA12 · June 2023 · Q7
 (a) Proven. (b) $\alpha = 4.491$. (c) H is a maximum at this value
- Q19** Pearson IAL WMA12 · October 2024 · Q4
 (a) $\frac{2}{\sqrt{x}} - \frac{9}{2x^{3/2}}$. (b) $x = \frac{9}{4}$. (c)(i) $-\frac{1}{x^{3/2}} + \frac{27}{4x^{5/2}}$. (c)(ii) Minimum. (d) $0 < x < \frac{9}{4}$

IDEA 03 · OPTIMISATION: BUILD THE FORMULA, THEN DIFFERENTIATE

- Q20** Dr Eslam · A box
 $A' = 2x - \frac{2000}{x^2} = 0: x = 10; A = 300 \text{ cm}^2$
 $(A'' = 6 > 0)$
- Q21** Practice · Medium
 $(a) h = \frac{150}{r^2}, S = 2\pi r^2 + 2\pi rh$ (b) $r = \sqrt[3]{75}$:
 $S_{\min} = 335.23 \text{ cm}^2$
- Q22** Practice · Very Hard
 $(a) OP^2 = x^2 + (3 - \frac{1}{4}x^2)^2$ (b) $\frac{d}{dx}OP^2 = 0$ at $x = \pm 2$
(second derivative > 0): $OP_{\min} = 2\sqrt{2}$
- Q23** Practice · Very Hard
 $(a) \theta < 2\pi$ so $A = \frac{1}{2}r^2\theta < \pi r^2$ (b) $P = 2r + r\theta = 2r + \frac{2A}{r}$ (c) $P' = 0$ at $r = \sqrt{A}$: $P = 4\sqrt{A}$; $P'' = \frac{4A}{r^3} > 0$
- Q24** Pearson IAL WMA12 · June 2021 · Q9
 (a) Proven: $S = \frac{250000}{h} + 2000\sqrt{h}$. (b) $h = 250^{2/3}$, so $k = \frac{2}{3}$. (c) Minimum
- Q25** Pearson IAL WMA12 · January 2023 · Q2
 $(a) S = 6x^2 + \frac{2592}{x}$. (b) $\frac{dS}{dx} = 12x - \frac{2592}{x^2}$. (c) $x = 6$. (d) Minimum shown.
 $(e) 648 \text{ cm}^2$
- Q26** Pearson IAL WMA12 · January 2025 · Q3
 $(a) A = 3x^2 + \frac{320}{x}$, so $P = 3$ and $Q = 320$. (b)
 $x = 3.76$. (c) $\frac{d^2A}{dx^2} = 6 + \frac{640}{x^3}$, minimum
- Q27** Pearson IAL WMA12 · June 2025 · Q9
 $(a) S = 6x^2 + \frac{20}{x}$, so $P = 6$ and $Q = 20$. (b)
25.3. (c) Minimum justified
- Q28** Pearson IAL WMA12 · June 2026 · Q9
 $(a) r = \frac{x-10}{2}: xy + \frac{\pi}{2}r^2$ (b) y from $A = 10000$
(c) $P' = 0: x = 84.6$ (d) $P'' = \frac{2(20000-25\pi)}{x^3} > 0$:
minimum

IDEA 04 · STATIONARY POINTS WITH AREAS

- Q29** Dr Eslam · Then the area $(2, 4)$ max; $\int_0^4 (4x - x^2) dx = \frac{32}{3}$
- Q30** Pearson IAL WMA12 · October 2021 · Q8
 (a) Proven: $k = 28$. (b) $x < 2$ or $x > \frac{7}{2}$. (c) $\frac{40}{3}$
- Q31** Pearson IAL WMA12 · October 2023 · Q9
 (a) y is increasing for $x > \frac{9}{4}$. (b) Area = $\frac{857}{21}$
- Q32** Pearson IAL WMA12 · January 2024 · Q10
 (a) Proven. (b) $\frac{373}{3}$
- Q33** Pearson IAL WMA12 · June 2024 · Q9
 (a) $x = \frac{12}{5}$. (b) $k = \frac{28}{5}$
- Q34** Pearson IAL WMA12 · June (R paper) 2024 · Q10
 (a) $x = 3$. (b) $\frac{148}{5}$
- Q35** Pearson IAL WMA12 · January 2026 · Q10
 (a) $x = 2\sqrt{5}$. (b) $k = \frac{10\sqrt{21}}{3}$